

Step	Step Description	Agents Involved	Tool Calls and Parameters	Reason
1	User requests all special education needs school sites in Exeter	human, host_agent	Message from human to host_agent: "Show me all the special education needs school sites in Exeter."	User initiates the query; host_agent receives and routes the request.
2	Plan steps to solve the query	planning_agent	send_message to planning_agent with task_description: "User wants to see all special education needs school sites in Exeter. Provide the general steps to solve this geospatial query."	planning_agent is chosen to decompose the query into actionable steps, as per its system prompt.
3	Output planning steps	planning_agent	Output: ["Find Exeter 1 area", "Search for special education needs school sites in Exeter as many search results"]	Steps are based on the prompt and reasoning: first, find the area polygon for Exeter, then search for the relevant school sites within it.
4	Generate metadata for all artifacts	host_agent	generate_metadata_for_all_artifacts	Ensures awareness of available data artifacts before proceeding.
5	Find area polygon for Exeter	named_area_agent	send_message to named_area with task_description: "Find the area polygon for Exeter."	named_area_agent is chosen because the prompt specifies that named_area contains polygons of areas like cities.
6	Query named area database for Exeter polygons	named_area_agent	call_os_ngd with polygon_or_point=True, filename='exeter_area_polygon'	The named_area_agent uses the NGD tool to retrieve all named area polygons, as per its API capabilities.
7	Filter to the specific Exeter polygon	data_analysis_agent	send_message to data_analysis_agent: "Filter the artifact 'exeter_area_polygon' to find the specific polygon for Exeter. Save the filtered results as 'filtered_exeter_area_polygon'."	The initial NGD query returns many polygons; further filtering is needed to isolate the one for Exeter.
8	Generate metadata for 'exeter_area_polygon'	data_analysis_agent	generate_metadata_for_artifacts with artifact_names=['exeter_area_polygon']	Metadata is generated to understand the columns and structure for accurate filtering.
9	Execute code to filter Exeter polygon	coding_agent, code_executor	code_executor with code that filters for rows where name1_text contains 'Exeter' and descriptiongroup/description indicate a settlement/city/town/suburban/village	The code uses 'name1_text' and relevant description columns, as indicated by the metadata, to ensure only the correct Exeter polygon is selected.
10	Confirm filtered Exeter polygon artifact	data_analysis_agent	Output: Filtered to 1 polygon for Exeter; artifact 'filtered_exeter_area_polygon' generated	Confirms that the correct polygon for Exeter has been isolated for further spatial queries.
11	Search for school sites within Exeter polygon	land_use_features_agent	send_message to land_use_features with task_description: "Search for special education needs school sites within the artifact 'filtered_exeter_area_polygon'."	land_use_features_agent is chosen because its prompt covers searching for land use features like schools within a given area.
12	Query land use features for schools in Exeter	land_use_features_agent	call_os_ngd with filters=['School'], bbox='filtered_exeter_area_polygon', filename='special_education_needs_schools_exeter'	The agent uses the 'School' filter and the Exeter polygon as the bounding box, as per the API and prompt.
13	Filter for special education needs schools	data_analysis_agent	send_message to data_analysis_agent: "Filter the dataset 'landuse_special_education_needs_schools_exeter' to identify special education needs school sites. Save the filtered results as 'filtered_special_education_needs_schools_exeter'. Include filtering using the 'oslandusetierb' column."	The initial search returns all schools; further filtering is needed using the 'oslandusetierb' column, as recommended in the land_use_features prompt and metadata.
14	Generate metadata for school sites artifact	data_analysis_agent	generate_metadata_for_artifacts with artifact_names=['landuse_special_education_needs_schools_exeter']	Metadata is generated to confirm the presence and meaning of the 'oslandusetierb' column for filtering.
15	Execute code to filter special education needs schools	coding_agent, code_executor	code_executor with code that filters rows where 'oslandusetierb' contains 'Special Needs' (case-insensitive)	The code uses the 'oslandusetierb' column, as indicated by the metadata and prompt, to identify special education needs schools.
16	Confirm filtered special education needs schools artifact	data_analysis_agent	Output: 7 entries found; artifact 'filtered_special_education_needs_schools_exeter' generated	Confirms that the correct subset of schools has been isolated.
17	Plot the special education needs schools in Exeter	plotting_agent	send_message to plotting_agent: "Plot the artifact 'filtered_special_education_needs_schools_exeter' to show all special education needs school sites in Exeter."	plotting_agent is chosen to visualize the results, as per its system prompt.
18	Generate metadata for plotting	plotting_agent	generate_metadata_for_artifacts with artifact_names=['filtered_special_education_needs_schools_exeter']	Metadata is generated to understand the structure for plotting.
19	Execute code to plot schools on a map	coding_agent, code_executor	code_executor with code that creates a folium map, plots polygons for each school, adds tooltips with name and area, saves as HTML	The code uses the geometry and name columns, as indicated by the metadata, to create an interactive map.
20	Confirm map artifact and summarize results	plotting_agent	Output: Map with 7 schools plotted, artifact 'special_education_needs_schools_exeter_map' generated	Confirms that the visualization is complete and provides a summary of the results.
21	Final summary to user	host_agent	Output: All special education needs school sites in Exeter have been found and plotted; 7 sites identified and mapped	host_agent summarizes the entire workflow and results for the user.

Overall Summary: - Databases used: Named area database (to find Exeter polygon), Land use features database (to find schools), and plotting tools for visualization. - Filters/code: Filtering was performed on the named area data using 'name1_text' and settlement/city/town/suburban/village descriptions to isolate Exeter. The land use data was filtered using the 'oslandusetierb' column to identify 'School For Special Needs'. - Final outcome: 7 special education needs school sites in Exeter were identified and plotted on a folium map, with each site shown as a polygon and labeled with its name and area. - Overall reasoning: Each agent and tool was chosen based on its system prompt and the metadata of the artifacts. Filtering steps were guided by the authoritative column schema provided by the metadata, ensuring accurate and relevant results at each stage. The workflow followed a logical progression from area identification, entity search, attribute filtering, to final visualization.